



Characterizing the permeability of a foulant layer under a transient applied pressure

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ABSTRACT

Pressure-driven membrane filtration systems are commonly operated at constant pressure, under which the presence of polymeric foulants deposited on the membrane surface leads to a decline in permeate flux. In the absence of further deposition, the polymeric foulant layer would typically compact in response to the permeate flow, resulting in a constant hydraulic resistance and a steady-state permeate flux lower than that of the clean membrane. However, the transient permeability of these soft, deformable porous layers remains poorly understood. Using a customized microfluidic nanofiltration system, we experimentally investigate the transient behavior of a porous deformable film under pressure steps and pressure waveforms. We demonstrate that a membrane fouled with a hydrogel layer can exhibit a transient permeability shift immediately after a sudden change in applied pressure. Moreover, this transient state suggests that compaction and swelling of the soft porous layer follow two different processes, involving distinct relaxation times and permeate volumes during the transient window induced by pressure increases and decreases, respectively. In addition, by imposing pressure waveforms, we show that a non-linear flow response arises as a result of the asymmetric dynamics of swelling and compaction of the soft material. Lastly, we demonstrate that such observations can have practical implications, as the non-linear flow response to oscillatory pressure can, in some cases, generate an enhanced overall permeability when compared to the averaged value of the pressure periodic function.

1. Introduction

In membrane-based water filtration, the rejection of organic molecules can lead to the formation of a porous cake layer on the membrane surface, which increases the hydraulic resistance and operational costs. These cake layers are often a representative example of soft, deformable porous materials, whose thickness can vary dynamically in response to the stresses induced by permeate flow.

The mechanics of soft, deformable porous materials are relevant to a broad scope of disciplines, encompassing both natural and engineered systems, with practical applications in chemical engineering, water filtration, food engineering, biotechnology, and soil mechanics. An additional layer of complexity arises when the compaction of the cake layer reduces its porosity and further increases hydraulic resistance.

The flow-induced compaction of a porous soft material is a particular case of poroelasticity [1], where the material's solid skeleton sustains the frictional stresses generated by the permeate flow, leading to compaction [2]. The coupling between interstitial flow and compaction of the porous solid skeleton leads to thickness changes in the soft material when varying the pressure, as the stresses generated by

interstitial flow cause a reduction in the void fraction within the porous structure.

The time scales for compaction and swelling in soft materials can differ, with some studies indicating that compaction may generally occur more rapidly than swelling [3]. When these materials face positive sinusoidal pressures, they might show frequency-dependent shifts in permeate flow relative to the applied pressure [4]. However, the transient permeability characteristics during these dynamic states remain largely unexplored.

Previous studies on transmembrane pressure-pulsing have shown that average filtration performance can be improved by maintaining a stable TMP with brief pulses of negative pressure applied at specific intervals [5]. The principle of this technique is to take advantage of the transient development of the concentration polarization resistance [6]. This approach opened the space for high-frequency backpulsing, where quick negative pulses of pressure not only aim to dissipate the concentration polarization but also to prevent the build up of foulant cakes on the membrane surface [7].

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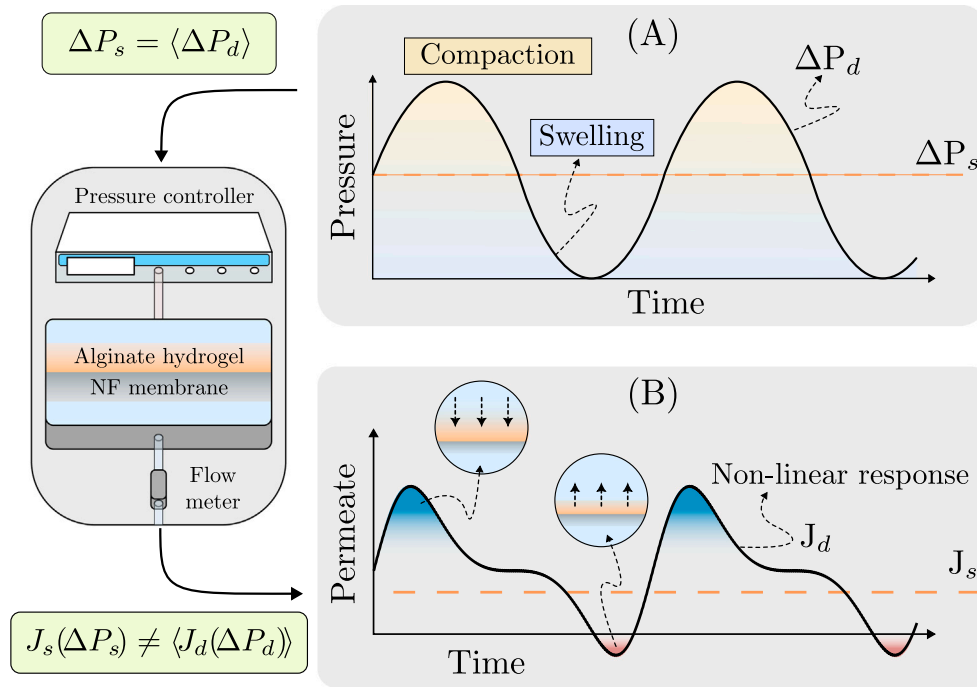


Fig. 1. Schematic illustration comparing the conceptual effects of a time-periodic applied pressure, driving water through an organic foulant layer, with its analogous case — static pressure (ΔP_s). The average value of the dynamic pressure ($\langle \Delta P_d \rangle$) is equal to the static pressure. The porous, deformable, foulant can undergo cycles of compaction and swelling in response to the pressure waveform. (B) The alternating cycles of compaction and swelling can result in a non-linear permeate flux response over the pressure waveform cycles, leading to different performances when compared to the analogous static pressure ($\langle J_s \rangle \neq \langle J_d \rangle$). This effect is explored throughout this study.

Additionally, osmotic backwashing presents an alternative to conventional backwashing methods that rely on applying negative pressures. Physically, when the transmembrane pressure drops below the osmotic pressure difference ($\Delta\pi$), water from the permeate side begins to diffuse back through the RO membrane [8,9]. Osmotic backwashing is a time-dependent process, characterized by an initial high water flow driven by the elevated osmotic pressure of the concentration polarization (CP) layer. This transient flow gradually decreases as the osmotic gradient decreases and the CP layer dissipates [10].

We hypothesize that modifying the transmembrane pressure (TMP) within the positive pressure range, i.e., without applying a negative back-pressure, could also influence transient permeability, particularly in cases where the membrane is fouled with a polymeric foulant carrying a net charge. In such cases, beyond the effects on concentration polarization, the polymeric foulant would undergo cycles of compaction and swelling in response to increasing and decreasing pressure, respectively.

Specifically, the time-varying pressure waveforms may alter the overall filtration performance by inducing repeated cycles of compaction and swelling in the foulant layer. These dynamic structural changes created by a pressure waveform could lead to different overall permeability, compared with the analogous static pressure condition, as illustrated in Fig. 1. The time-averaged permeate flux over n cycles of a periodic pressure signal with period T is given by

$$\langle J_d \rangle = \frac{1}{nT} \int_0^{nT} \frac{\Delta P_d(t) - \Delta\pi^*(t)}{\mu[R_m + R_c(t)]} dt,$$

where $\langle J_d \rangle$ is the time-averaged permeate flux under dynamic pressure, ΔP_d is the time-varying applied pressure, which represents the pressure difference between the feed and the permeate side of the membrane, also known as trans-membrane pressure (TMP), μ is the solution viscosity, $\Delta\pi^*$ is the cake-enhanced osmotic pressure [11], and R_m and R_c are the hydraulic resistances of the membrane and the foulant cake, respectively.

In this manuscript, we show that, in addition to the known phase shift between an applied dynamic pressure and the resulting permeate flux in fouled membranes [4,12] (see Fig. 1), a transient permeate flux can also emerge in response to sudden pressure changes. Furthermore, we demonstrate that the dynamic permeability exhibits a non-linear response to the pressure waveform. We investigate how different waveform shapes influence this transient permeability and, consequently, affect overall filtration performance.

2. Materials and methods

The experimental setup is based on a custom-made nanofiltration cell [13,14], monitored by a programmable pressure-controller (Fluigent MFCS-EZ) connected to a microfluidic flow sensor (Fluigent FRP), so that the permeation flux is continuously monitored, while the applied pressure is modulated. The model foulant used is a calcium alginate, which creates a hydrogel film at the membrane surface. The membrane employed for the experiments is a nanofiltration membrane (NF270, Dupont) with an effective filtration area of 2.16 cm² (dimensions: 0.6 mm (H) × 6 mm (W) × 36 mm (L)).

Alginic acid sodium salt (sourced from brown algae, Sigma), was dissolved in deionized water to a concentration of 25 g/L and placed on a shaker overnight to ensure complete dissolution and homogeneity. At the beginning of each experiment, 0.6 mL of the solution was pipetted directly into the feed channel of the filtration cell. The system was then operated at a pressure of 1 bar for 30 mins using deionized water, allowing for even coverage and uniform formation of the fouling layer on the membrane surface. Since the foulant in this system is an ionically-crosslinked alginate layer, the purpose of this deposition step is to prevent the cross-linking of the alginate with calcium ions from occurring during the deposition, to avoid the formation of aggregates and ensure a uniform layer. Once the membrane becomes fouled with alginate, the DI water feed is replaced with one containing calcium ions to form the final calcium alginate layer. The feed solution used

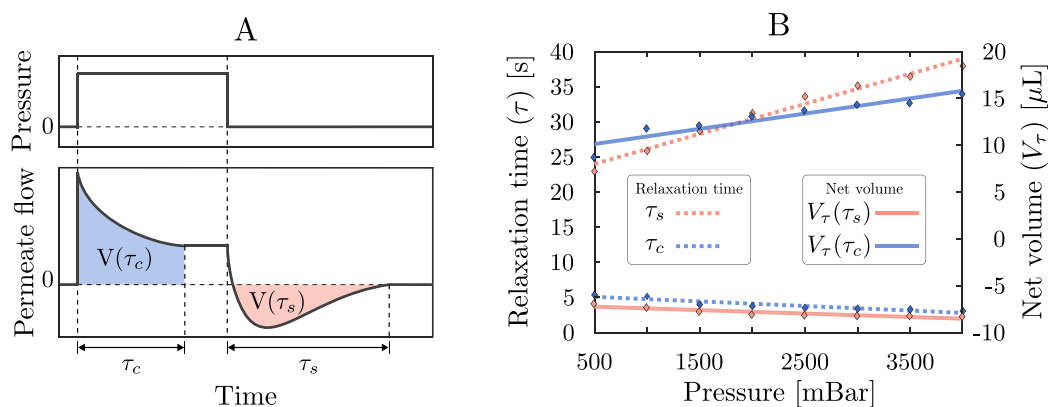


Fig. 2. The transient properties of the studied system, i.e. the relaxation times, are examined by applying pressure steps. (A) The transient permeate flow response to a sudden pressure increase followed by a pressure release is shown. Upon increasing the pressure, the permeate flow initially rises before gradually decreasing as the foulant layer compacts. Swelling occurs when the pressure is released. Additionally, during the pressure release, the transient permeate flow reverses due to the osmotic driving force. The relaxation time (τ) is defined as the time required for the soft material to reach a steady state following a sudden pressure change. The net permeated volumes ($V_\tau(\tau_c)$ and $V_\tau(\tau_s)$) represent the volume of permeate during the transient phases of compaction and swelling. (B) The relaxation times (τ_c for compaction and τ_s for swelling) are plotted as a function of pressure, along with the corresponding net permeated volumes ($V_\tau(\tau_c)$ and $V_\tau(\tau_s)$).

throughout the experiments was a 1 mM aqueous solution of anhydrous calcium chloride (Baker). Prior to starting the experiments, the system was operated using the calcium solution at a pressure of 1 bar for, at least, one hour to ensure it reached an apparent steady state or steady permeate flow.

It is important to note that all pressure variation experiments were conducted with a pre-formed alginate fouling layer, and no alginate was present in the feed during these measurements. Only the 1 mM CaCl_2 solution was used after the fouling layer had been formed. Thus, no additional foulant was introduced during the experiments.

Two types of experiments were carried out. First, relaxation experiments, which involved imposing pressure steps on the hydrogel sample to study its transient response. Second, periodic pressure experiments, which tested the permeation under three waveforms: sinusoidal, positive ramp, and negative ramp. For each waveform type, time-based protocols were employed, beginning with the lowest amplitude and testing both periods ($T = 64$ s and $T = 32$ s). The amplitudes were systematically varied from the lowest (500 mBar) to the highest tested values. In this work, we use the term ‘relaxation’ in broad reference to the transient response of the soft fouling layer following a step change in pressure, including both compaction under increased pressure and swelling upon pressure release.

The pressure waveforms were generated using a microfluidic pressure controller (Fluigent MFCS-EZ), integrated with a flow sensor (Fluigent FRP). The waveform profiles were programmed and regulated through the OxyGEN software (Fluigent). After each experiment, the resulting datasets were analyzed using MATLAB to evaluate and interpret the findings. All pressure variation experiments were conducted in a dead-end configuration, without cross-flow.

3. Results and discussion

3.1. Dynamics of relaxation

In this section, we assess the dynamics of compaction and swelling of the alginate layer, which serves as a model soft porous material, in order to gain insight into the time-scales involved. The system’s transient behavior was characterized through pressure-step experiments. Each test began at 0 bar, followed by a step increase to a set pressure and a subsequent release back to 0 bar, with one-minute intervals between each step (see 2A). Following each pressure increase, the permeate flux was monitored until it reached an apparent steady state, operationally defined as a period of constant flux. Upon pressure release, the subsequent time-dependent recovery of the flux was also

recorded. While we define a relaxation time, τ , as the time required for the flux to stabilize after each step, we note that this definition is operational and specific to the observed system. It serves as a practical metric for comparing relative response times across conditions in the further analyzed dynamic response, rather than a universal material property.

The system’s permeability and response under the pressure-stepping, until a steady-state is reached, provides insights that is then useful for understanding the dynamics under varying pressure waveforms. The results revealed a remarkable difference in the net water volume displaced during the period of increasing and decreasing pressure (see Fig. 2). Additional details of the pressure-step procedure are provided in Figs. S1 and S2, while an expanded presentation of the results shown in Fig. 2 can be found in Figs. S3 and S4 of the Supplementary Information.

The pressure step analysis confirmed that the fouled membrane exhibits time-dependent permeability, reaching an apparent steady-state within a few seconds (see Fig. 2.B). This steady-state is attained more quickly during the positive permeation phase (when pressure is applied, causing compaction) than in the reverse case, where pressure is reduced to zero, allowing swelling to occur. These findings align with previous research on the compaction and relaxation of biofouling layers on membrane surfaces, where transient deformation of the soft material structure was also observed within a similar time frame [3]. A similar trend has been reported in processes involving the swelling and deswelling of hydrogels, where water uptake occurs faster than the deswelling process [15].

We define the compaction relaxation time (τ_c) as the time required for the system to reach an apparent steady flowrate after a sudden increase of pressure. The results indicate that increasing the pressure steps slightly decreases τ_c , see 2.B. Further, the relaxation time is 5 s at the lowest pressure of 500 mBar, and 3 s for the highest analyzed step pressure of 4000 mBar, see Fig. 2.B. The total permeated volume of water (V_τ) increases for the compaction steps (positive applied pressure) from 8 μL to 15 μL at pressure steps varying from 500 mBar to 4000 mBar, respectively, as shown in Fig. 2. On the other hand, the ‘negative’ volume, displaced during the transient de-compaction (upon reduction of the applied pressure to 0 mBar) is less sensitive to the pressure steps, varying only from -6 μL to -8 μL over the same range. In contrast to τ_c , the pressure release step exhibits an increase in relaxation time (the swelling time-scale, τ_s), from 22 s at a pressure step of 500 mBar to 38 s at 4000 mBar.

The compaction dynamics of a soft porous material, examined through positive steps and pressure release steps, indicate that compaction and swelling correspond to positive and negative net filtered

volumes, respectively, see Fig. 2.B. This discrepancy aligns with the distinct mechanisms governing swelling and compaction in soft porous materials. In a hydrogel, compaction is flow-induced and results from permeation generated by applying a pressure gradient across the membrane [16–18]. In contrast, swelling occurs due to osmotic gradients between the gel polymeric network and the bulk fluid [15].

While the mechanisms involved in the compaction and swelling of the studied foulant layer cannot be fully resolved from the present measurements, it is important to note that the NF membrane used in this study partially rejects Ca^{2+} ions from the feed solution. These divalent cations can interact with the carboxyl groups of the guluronic acid-rich segments, known as G-blocks, in the studied alginate film, forming a reversible non-covalent bond [19]. Such interactions may influence the mechanical behavior of the fouling layer, potentially contributing to its resistance to deformation and affecting its relaxation dynamics, due to a simultaneous evolution of cake-enhanced concentration polarization of the rejected Ca^{2+} ions within the foulant. Although the bulk osmotic pressure difference ($\Delta\pi$) between the feed and permeate is negligible at 1 mM CaCl_2 , we hypothesize that the fouling layer may act as a ‘capacitive’ reservoir for Ca^{2+} ions, concentrating them locally through ionic complexation. This localized ion enrichment near the membrane surface could modify the effective osmotic conditions within the foulant layer, potentially contributing to permeate backflow during pressure release. A detailed investigation of this local osmotic modulation and its impact on relaxation dynamics is a promising direction for future work.

Returning to the experimental findings, for each pressure step, the net permeate volume during the relaxation times of compaction, $V_c(\tau_c)$, and swelling, $V_s(\tau_s)$, differ in magnitude (see Fig. 2.B). Furthermore, for the analyzed pressure steps, the estimated rate of change of $V_c(\tau_c)$ with pressure is 1.6 $\mu\text{L}/\text{bar}$, whereas for $V_s(\tau_s)$, it is $-0.3 \mu\text{L}/\text{bar}$. This asymmetry suggests that filtration performance could potentially be optimized by leveraging the distinct responses of the material to compaction and swelling.

3.2. Permeation response to applied pressure modulation

Next, we examine the transient permeability of the foulant layer by employing 3 pressure waveforms: sinusoidal, as well as both negative and positive ramps: in the positive ramp waveform the pressure increases linearly to its maximum value, whereby it drops abruptly to 0, while the negative ramp rises abruptly to the maximum value, and then decreases linearly to 0. For each waveform, the applied pressure amplitudes are 500, 1000, 1500, and 2000 mBar. To ensure all pressures remain non-negative, the waveforms are vertically shifted according to their respective pressure amplitudes, making 0 mBar the lowest value in each cycle.

As may be seen in Fig. 3, the response of the permeation to the pressure waveforms was nonlinear, with a clearly exhibited hysteresis. Moreover, the three types of pressure waveforms resulted in different shapes of the obtained hysteresis curves, at identical values of applied pressure amplitude. In all cases, a region of negative flow rate is observed, as seen in Fig. 3. In contrast to the pressure release steps (shown in the previous section), the pressure and flow-rate exhibited a phase shift when the system was subjected to a pressure waveform, see Fig. 3. This effect is consistent with findings in the literature on so-called ‘hydraulic impedance spectroscopy’ [12].

Obvious differences are observed in the hysteresis curve of pressure and flux across the different pressure waveforms studied, demonstrating how the transient permeability is linked to deformation dynamics. For the sinusoidal pressure waveform (Fig. 3.I), the highest permeation rate is observed as the pressure increases. Additionally, as the pressure decreases, the flow rate appears to decrease linearly — meaning that the foulant layer relaxes faster than the rate of pressure change. In the case of the positive ramp waveform, which induces abrupt decreases in pressure, the curve shows a discontinuity after reaching its highest

pressure values (see Fig. 3.II). Lastly, the negative ramp waveform displays the highest flow rate values, as the pressure rises abruptly in this case (see Fig. 3.III). This sudden increase in pressure and flow rate likely leads to rapid compaction of the foulant layer, which probably causes the transient flow rate to reach higher values compared to the sinusoidal and positive ramp waveforms. Time-domain representations of pressure and permeate flux for each pressure waveform are provided in Figs. S5–S8 of the Supplementary Information. Figs. S9–S15 present comparisons across different periods at fixed amplitude and waveform type, together with the corresponding static pressure reference.

Although the results for pressure vs. flow rate indicate that the negative ramp waveform can achieve the highest permeation rates compared to the other waveforms, this is not necessarily indicative of better filtration performance. The impact of the pressure wave on filtration performance is not straightforward, due to the coupling between the permeation and the flow-induced compaction in the foulant layer [20]; Indeed, a higher permeation rate may accelerate the compaction of the porous layer, reducing its permeability. In the upcoming section, we assess the filtration performance of the different waveforms by comparing the time-averaged permeation volume against the equivalent static pressure conditions.

3.3. Filtration performance

In order to assess the filtration performance, we turn to analyze the overall permeate volume obtained under dynamic and static applied pressure. For the dynamic system, all pressure waveforms are shifted by half their amplitude to ensure that the pressure varies between zero and the designated positive values. To facilitate a meaningful comparison between static and dynamic conditions, the static pressure is then adjusted to match the average value of the dynamic pressure at a given amplitude, thereby ensuring a uniform reference point. This ensures that the time-averaged applied pressures in both cases are equal, i.e., $\langle \Delta P_s \rangle = \langle \Delta P_d \rangle$.

The results indicate a difference in the filtration performance under static and dynamic conditions. First, Fig. 4.A presents the case of a dynamic pressure applied as a sine wave at an amplitude of 500 mBar and periods of 64 and 32 s. In this configuration, the performance of the pressure sine wave yielded values below the static pressure, specifically -19.2% and -20.8% for the periods of 64 and 32 s, respectively (see Fig. 5). Furthermore, comparing the negative ramp pressure waveform with the static pressure of 500 mBar, resulted in performance reductions of -16.9% and -27.4% , for periods of 64 and 32 s, respectively. In particular, the value of -27.4% represents the lowest performance among all tests (see Fig. 5). Additional statistical analyses over repeated cycles are presented in Fig. S16 of the Supplementary Information. Indeed, the reduced performance of the negative ramp pressure results from a mismatch with the relaxation times observed for the system during positive and negative pressure steps. The negative ramp, characterized by a sudden pressure increase followed by a linear decrease, does not effectively leverage the net volumes displaced by the transient states. As observed in Fig. 3.III, the negative ramp waveform achieves the highest flow rates of the studied waveforms. The filtration results suggest, in a manner reminiscent of the Icarus myth, that attaining the highest flow rates may accelerate the compaction of the porous layer, producing a precipitous decline in subsequent transient flow rates.

In contrast, when employing a positive ramp waveform and an amplitude of 500 mBar and 64 s period, performance increased slightly, by 2.8%, above the static pressure. Performance further increased at the higher frequency (the period of 32 s), at 24.2% above the static pressure (see Fig. 5), demonstrating the highest performance among the examined waveforms. This enhancement can be attributed to the effective exploitation of the positive net filtered volume during the transient state, as this waveform aligns with the relaxation times and the displaced volume, throughout each phase.

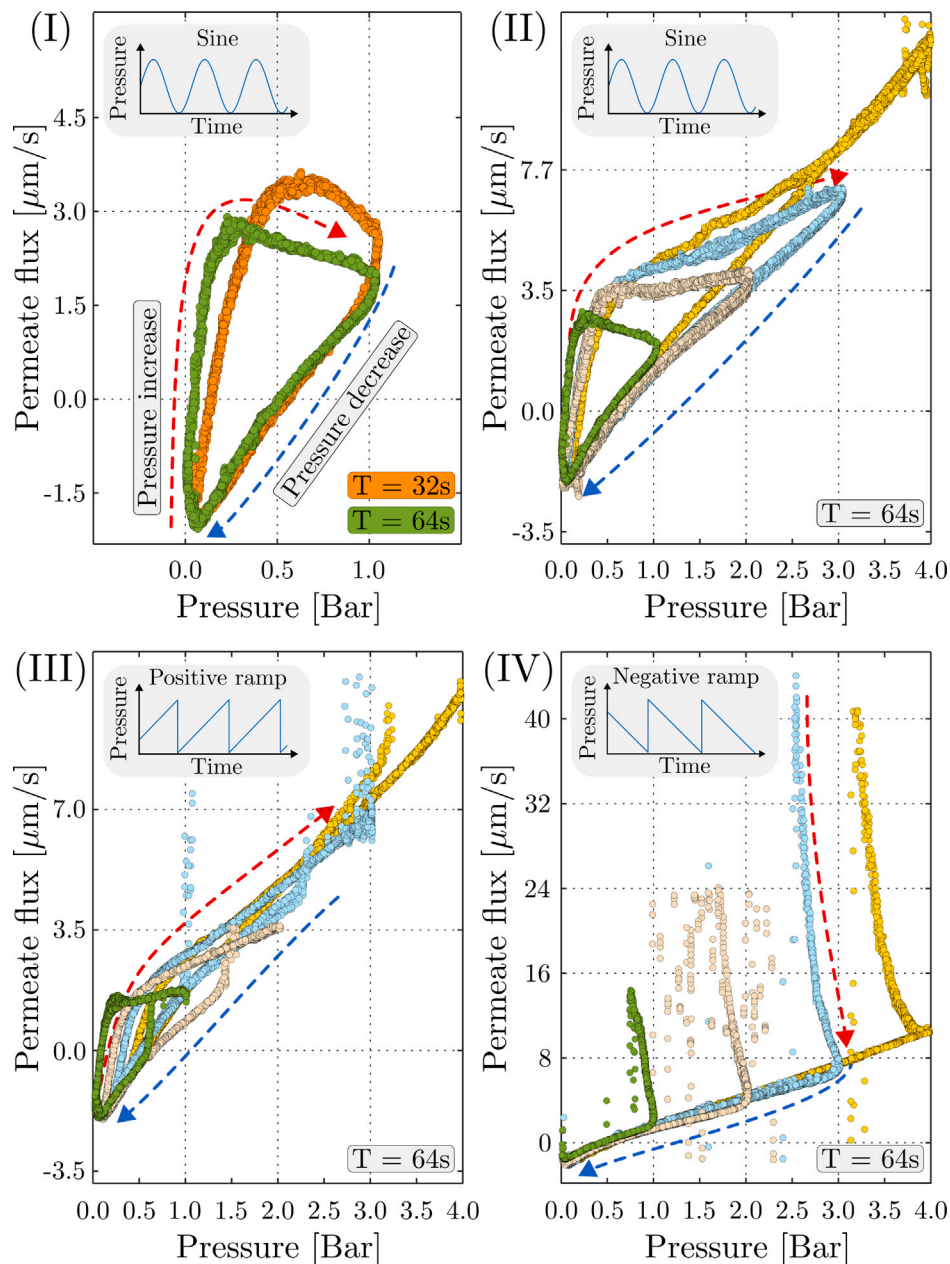


Fig. 3. Pressure versus flow-rate plots showing the hysteresis behavior of a soft porous material under different pressure waveforms. (I) Flow-rate response to sinusoidal pressure waves with two different pressure periods ($T = 32\text{ s}$ and $T = 64\text{ s}$). (II–IV) Flow-rate response to pressure waveforms of period of ($T = 64\text{ s}$) while varying their amplitudes by 500, 1000, 1500, and 2000 mBar. The phases of pressure increase and decrease, are indicated as a red dashed line and blue dashed line, respectively. The hysteresis loops illustrate the material's transient response, characterized by the differences in flow rate during the pressure cycles. Particularly, the amplitude analysis is shown for the following pressure waveforms: (II) Sinusoidal, (III) Positive ramp, and (IV) Negative ramp.

The filtration performance of the studied waveforms can also be assessed by the individual contributions of each cycle. Comparing the performance over two periods indicates that shortening the period may amplify the positive or negative effect, as we observed the lowest performance for the negative ramp (−27.4%) and the highest for the positive ramp (24.2%) for the shortest period examined (32 s).

The increase in pressure amplitude shows an overall improvement of filtration performance, see Fig. 5. This improvement can be linked to the observations from the pressure-step analysis, where the net filtered volume (V_f) increases with positive pressure steps, see Fig. 2.B. Here, the pressure waveforms may find an improvement in their performance as this positive filtered volume increases. In addition, the relaxation time (τ), also obtained from the pressure-step analysis, appears to be reflected in the variations of filtration performance

observed between the different studied pressure waveforms. Specifically, the studied waveforms exhibit three distinct relaxation behaviors corresponding to the compaction and swelling dynamics of the foulant cake. A direct measurement of the foulant cake deformation, along with its interaction with concentration polarization, would be necessary to better understand the transient permeability of the system.

4. Concluding remarks

The presented study delves into the dynamic behavior of a foulant layer – a soft deformable porous material – under varying pressure and flow conditions, with the goal of shedding light on flow-induced compaction dynamics and its impact on filtration performance. Notably,

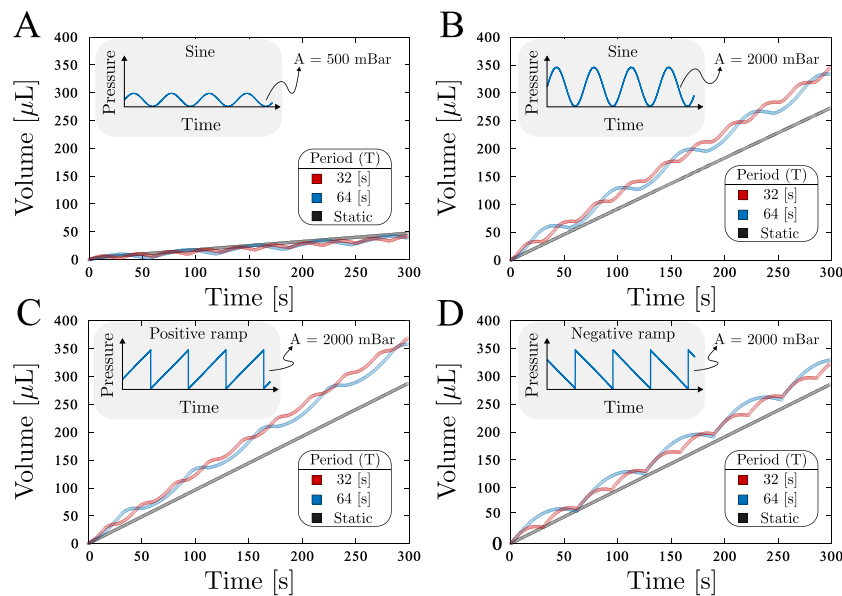


Fig. 4. The filtered volume over time for the studied system is displayed under static pressures and oscillatory pressures at periods of 32 and 64 s. Each measurement is represented by a circle on the plots: data in black indicate static pressure measurements, while blue and red data correspond to oscillatory pressures with periods of 32 s and 64 s, respectively. The figures present the following experimental conditions: (A) A 500 mBar sine pressure waveform. (B) A 2000 mBar sine pressure waveform. (C) A 2000 mBar positive ramp pressure waveform. (D) A 2000 mBar negative ramp pressure waveform.

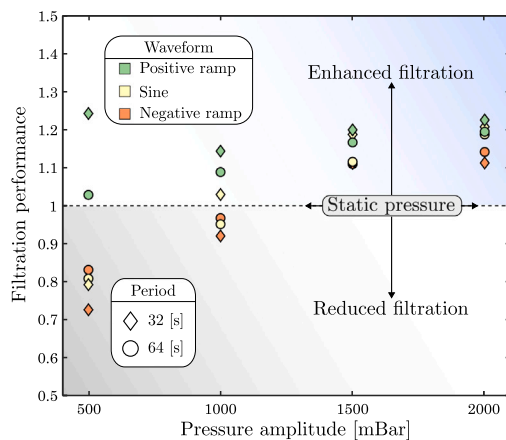


Fig. 5. The filtration performance of the studied waveforms is compared with that of their corresponding static pressures. The filtration performance of each waveform is determined by comparing the period-averaged permeate to the permeate achieved under the corresponding static pressure for the same period. The horizontal line at the unit value represents the filtration performance of the static pressure. The results indicate that the positive ramp waveform exhibits the highest filtration performance among the waveforms analyzed.

the observed asymmetry in relaxation times during compaction and decompaction, coupled with the distinct responses to different pressure waveforms, underscores the complexity of the material's behavior.

From a practical standpoint, the results indicate that pressure waves can be used to optimize the filtration processes, manifested in an increased permeation compared with the equivalent rate achievable with the corresponding average pressure. The results further highlight the importance of considering pressure wave characteristics, leveraged with respect to the characteristic time scales of compression and swelling — intrinsic properties of the foulant layer, requiring different waveforms to obtain optimal performance.

We observed a clear difference in the filtration performance between static and dynamic pressures in a system involving a highly

deformable soft porous material. This study indicates that a filtration unit fouled with a hydrogel cake deposited on the membrane can find enhanced performance through pressure waves. Additionally, pressure pulses exist in filtration processes in biological systems, which involve a semi-permeable surface and mucus as the soft porous material. The direction of this study may be relevant to further understanding filtration performance in these types of complex systems.

While this study provides valuable insights, the dynamic nature of the material prompts further exploration. Future research efforts should explore the mechanistic aspects of compaction, the influence of different pressure wave parameters, and evaluate the long-term response of the material under cyclic pressure conditions. Further research is required to analyze the filtration performance under dynamic pressures, especially employing periods below the relaxation time of the foulant layer. In particular, future work should further explore the conditions that lead to optimal performance of the dynamic pressure compared to the static pressure.

CRediT authorship contribution statement

José Agustín Epstein: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guy Z. Ramon:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124683>.

Data availability

Data will be made available on request.

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